Qualitative Research Methods in HCI

Human-Computer Interaction

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Why Qualitative Research?

Not all questions can be answered with numbers

Understanding users' experiences, attitudes, and behaviors

Discovering the "why" behind the "what"

Capturing rich, nuanced, contextual insights

Today's Journey

1. **Understanding Qualitative Research** — Philosophy, strengths, and when to use it
2. **Interviews** — Types, conducting, and best practices
3. **Observations** — Field work, contextual inquiry, and ethnography
4. **Analyzing Qualitative Data** — Thematic analysis, affinity diagramming, coding
5. **Quality in Qualitative Research** — Validity, reliability, transparency, ethics

Goal

Learn to gather and analyze rich qualitative data to understand users' worlds from their perspective

What is Qualitative Research?

Qualitative vs. Quantitative

Qualitative Research

- Explores meaning, experiences, attitudes
- Open-ended, flexible approach
- Rich, detailed descriptions
- Small samples, in-depth study
- **Questions**: Why? How? What does it mean?
- **Goal**: Understanding and insight
- **Data**: Words, images, observations

Quantitative Research

- Measures variables, tests hypotheses
- Structured, predetermined approach
- Statistical patterns and relationships
- Large samples, breadth of study
- **Questions**: How many? How much? How often?
- **Goal**: Generalization and prediction
- **Data**: Numbers, measurements

> Both are valuable! Often used together for comprehensive understanding

When to Use Qualitative Research

Qualitative research is ideal for:

- Early exploration of a new domain or technology
- Understanding user needs, goals, and pain points
- Investigating **why** users behave in certain ways
- Exploring attitudes, beliefs, and emotions
- Understanding context and real-world use
- Discovering unexpected insights

Examples in HCI:

- “How do people manage passwords in practice?”
- “What challenges do users face with voice assistants?”
- “Why do people abandon mobile apps?”
- “How does social media affect grief and mourning?”

> Use qualitative methods when you need depth over breadth, understanding over measurement

Core Characteristics

1. **Naturalistic inquiry** — Studying people in their real contexts
2. **Holistic perspective** — Understanding the whole situation
3. **Participants' viewpoint** — Seeing the world through users' eyes
4. **Flexibility** — Adapting approach as understanding grows
5. **Rich description** — Detailed accounts of experiences and contexts
6. **Iterative process** — Analysis begins during data collection

> You're not testing a hypothesis — you're building understanding from the ground up

Interviews

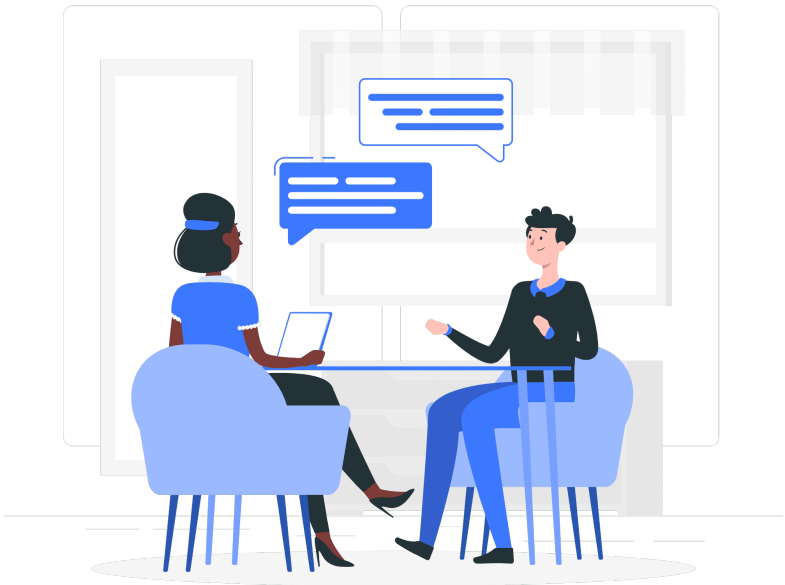
Why Interview Users?

Interviews help you understand:

- Users' experiences and perspectives
- Attitudes, beliefs, and motivations
- Behaviors in context
- Needs and pain points
- Stories and narratives

Not well-suited for:

- Statistical information ("How many times...?")
- Objective performance data
- Testing specific interactions
- Large-scale patterns



> Interviews reveal the "why" and "how" that you can't observe directly

Types of Interviews

1. Structured Interviews

- Fixed questions in predetermined order
- Same questions for all participants
- Can be quantitative (collecting statistics) or qualitative (exploring content)
- **Pros**: Easy to compare, reliable, efficient
- **Cons**: Less flexibility, may miss important topics
- **Use when**: You know exactly what to ask; comparability is important

Types of Interviews

2. Semi-Structured (Open-Ended) Interviews

- Loose structure with general topics
- Flexible question order and wording
- Can probe interesting responses
- **Pros**: Balance of structure and flexibility, rich data
- **Cons**: Requires skill, harder to compare
- **Use when**: Exploring experiences; balance needed

> Most common in HCI research

Types of Interviews

3. Unstructured Interviews

- No predetermined questions or order
- Conversation flows naturally
- **Pros**: Maximum flexibility, unexpected insights
- **Cons**: Difficult to analyze, requires expertise
- **Use when**: Exploratory, ethnographic work

Planning an Interview Study

1. **Define your research questions**
   - What do you want to understand?
   - Why are interviews the right method?
2. **Identify participants**
   - Who can answer your questions?
   - How many? (Usually 8-15 for interviews)
   - How will you recruit them?
3. **Develop interview guide**
   - Main topics/themes to cover
   - Example questions (for semi-structured)
   - Probes for following up
4. **Pilot your interview**
   - Test with colleague or friend
   - Refine questions and approach
   - Check timing (usually 45-90 minutes)

Conducting Effective Interviews

Before the interview:

- Prepare consent forms
- Test recording equipment
- Create comfortable environment
- Build rapport

During the interview:

- Start with easy, open questions
- Listen actively, don't interrupt
- Use probes: "Can you tell me more?"
- Follow interesting threads
- Watch for non-verbal cues
- Take brief notes (even if recording)

Key questioning techniques:

- **Open-ended**: "Tell me about..."
- **Probing**: "What happened next?"
- **Clarifying**: "What did you mean by...?"
- **Reflecting**: "So you felt..."

Avoid:

- Leading questions
- Yes/no questions (when possible)
- Multiple questions at once
- Jargon or technical terms
- Interrupting or filling silences too quickly

> Your goal: Create a space where participants feel comfortable sharing their experiences

Special Case: Contextual Inquiry

Contextual inquiry combines interviewing with observation in the user's environment

Principles

1. **Context** — Go to the user's workplace/environment
2. **Partnership** — Interviewer as apprentice, user as master
3. **Interpretation** — Create meaning together
4. **Focus** — Go for depth on key topics

The pattern:

- Observe user performing real tasks
- Ask questions about what they're doing
- Discuss their reasoning and decision-making
- Alternate between watching and talking (withdrawal and return)

Particularly valuable for understanding work practices and tool use

Examples from Research

Semi-Structured Interviews

- Voida, A., Harmon, E., & Al-Ani, B. (2011). Homebrew Databases: Complexities of Everyday Information Management in Nonprofit Organizations. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 915–924.

Discovered how nonprofits create “homebrew databases” (improvised systems cobbled together) that somehow work but are nearly impossible to maintain. Shows how interviews reveal the messy reality behind technology use.

Contextual Inquiry

- Raven, M. E., & Flanders, A. (1996). Using Contextual Inquiry to Learn About Your Audiences. *ACM SIGDOC Asterisk Journal of Computer Documentation*, *20*(1), 1–13.

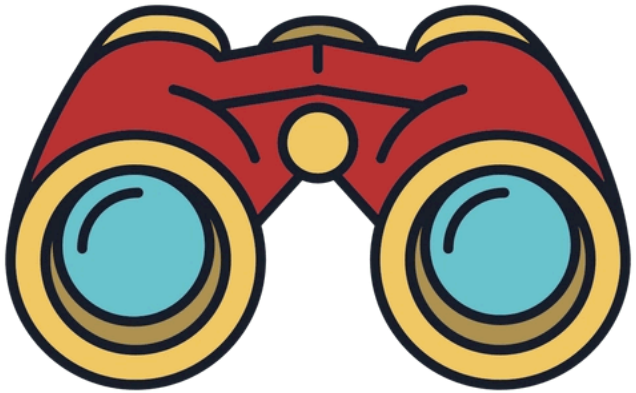
Researchers became “apprentices” to technical writers, watching them struggle with documentation systems during actual work. By observing in context rather than just asking questions, they discovered what users really needed versus what they said they needed.

Observations & Ethnography

Observational Research

Why observe users?

- People don't always do what they say they do
- Uncover tacit knowledge and habitual behaviors
- Understand social and environmental context
- See real practices, not idealized descriptions



Types of observation:

- **Direct observation** — Researcher present, taking notes
- **Video recording** — Detailed record for later analysis
- **Diary studies** — Users self-report observations
- **Ethnography** — Extended immersion in users' world

> Observation reveals what interviews might miss

Field Notes: Capturing Observations

What to record in field notes:

- What happened (activities, interactions)
- Who was involved
- When and where it occurred
- Physical environment and artifacts
- Social dynamics and relationships
- Your interpretations and questions
- Quotes and key phrases

Best practices:

- Write notes as soon as possible (details fade fast)
- Be descriptive, not just interpretive
- Note your own reactions and biases
- Use shorthand during observation, expand later
- Include sketches, diagrams, photos when appropriate

> Thick description: Detailed, rich accounts that convey the full context

Ethnography in HCI

Ethnography: Extended engagement with a community to understand their perspective

Four phases:

1. **Getting access** — Finding and approaching participants
2. **Developing membership** — Building trust, becoming familiar
3. **Collecting data** — Observations, interviews, artifacts
4. **Analyzing data** — Making sense, finding patterns

Key principles:

- Long-term engagement (months, even years)
- Holistic perspective
- Participant observation
- Cultural understanding

Ethnographic mindset:

- Approach as apprentice, open eyes
- Follow leads from participants
- Ask people to describe what they do
- Observe where outputs go
- Invest time in thinking, not just collecting
- Make everything strange (defamiliarization)

In HCI, used for:

- Understanding work practices
- Cultural perspectives on technology
- Communities and social interactions

Example from Research

Ethnography

- Heath, C., & Luff, P. (1992). Collaboration and Control: Crisis Management and Multimedia Technology in London Underground Line Control Rooms. *Computer Supported Cooperative Work (CSCW)*, *1*(1–2), 69–94.

Embedded in London Underground control rooms during crisis situations to understand how controllers coordinate trains and passengers during emergencies. Captured the split-second decisions, unspoken coordination, and tacit knowledge that keeps the system running — insights impossible to get from interviews alone.

Analyzing Qualitative Data

The Analysis Challenge

Qualitative analysis involves:

- **Expanding** — Exploring meanings and interpretations
- **Condensing** — Reducing data to patterns and themes

Two main approaches:

- **Inductive** (bottom-up) — Themes emerge from data
- **Deductive** (top-down) — Apply existing framework or theory

Often a combination of both!

> Analysis is iterative — you move back and forth between data and interpretation

Transcription: The First Step

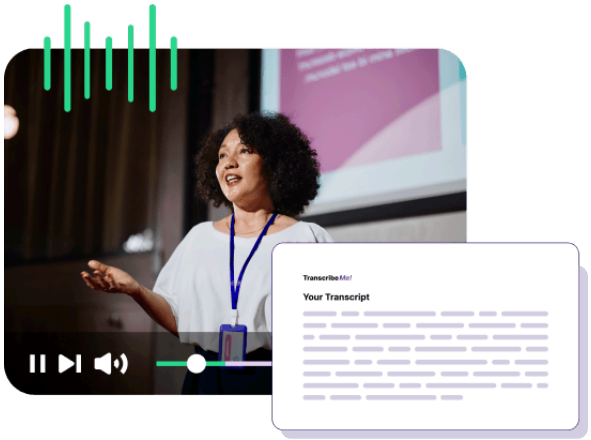
Transcription converts audio/video recordings to text

Level of detail varies:

- **Basic** — Just words, cleaned up
- **Verbatim** — Every word, pause, filler (“um”, “uh”)
- **Detailed** — Intonation, overlaps, timing

Practical considerations:

- 1 hour of audio ≈ 4-6 hours to transcribe
- Use transcription software to speed up
- Check accuracy — errors affect analysis
- Consider privacy (remove identifiers)



> Transcription is already analysis — you notice patterns as you type

Coding: The Foundation of Analysis

Coding: Assigning labels (codes) to segments of data

What codes capture:

- Activities or behaviors
- Emotions or attitudes
- Concepts or themes
- Processes or strategies
- Relationships or interactions

Coding can be:

- **Descriptive** — What's happening
- **Interpretive** — What it means
- **Pattern** — Types or categories

Example codes from interview about password use:

- "password reuse"
- "frustration with complexity requirements"
- "workaround: writing down passwords"
- "security/convenience tradeoff"
- "lack of understanding risk"

Codes range from specific (descriptive) to abstract (conceptual)

> Be generous initially — you can always collapse codes later

Method 1: Affinity Diagramming

Affinity diagramming (KJ Method): Collaborative, bottom-up grouping of data

Four phases:

1. **Create notes** — Break data into small, meaningful units (Post-its work well)
2. **Cluster notes** — Group related notes together, explain rationale
3. **Walk the wall** — Review organization, merge/split clusters, add/refine notes
4. **Document** — Transfer organization to digital format, name clusters

Strengths:

- Visual, tactile, collaborative
- Good for team sensemaking
- Relatively quick
- Surfaces themes naturally

> Particularly useful for workshop settings or team analysis

Method 2: Thematic Analysis

Thematic analysis: Systematic identification and organization of themes

Six steps:

1. **Familiarize** — Read transcripts, note initial thoughts
2. **Generate codes** — Label interesting features systematically
3. **Search for themes** — Group codes into potential themes
4. **Review themes** — Check themes against data, refine
5. **Define themes** — Name and describe each theme clearly
6. **Write up** — Weave themes into coherent narrative

> Gold standard for rigorous qualitative analysis in HCI

Method 2: Thematic Analysis

Thematic analysis: Systematic identification and organization of themes

What makes a good theme?

- Captures important pattern in data
- Relevant to research question
- Has substantial evidence (multiple instances)
- Coherent and internally consistent
- Distinct from other themes

> Gold standard for rigorous qualitative analysis in HCI

Method 2: Thematic Analysis

Thematic analysis: Systematic identification and organization of themes

Practical tips:

- Make multiple passes through data
- Keep memos of interpretations
- Use participants' own words for theme names
- Check themes with team or participants
- Show evidence (quotes) for each theme

> Gold standard for rigorous qualitative analysis in HCI

Coding Example

Transcript excerpt: "I use the same password for everything except my bank. I know I shouldn't, but I can't remember different passwords for every site. I tried a password manager once but it was too complicated."

Possible codes:

- `password_reuse` — Uses same password across sites
- `security_awareness` — Knows practice isn't ideal
- `memory_burden` — Difficulty remembering multiple passwords
- `exception_for_high_stakes` — Different password for bank
- `tool_rejection` — Tried password manager, found too complex
- `convenience_priority` — Prioritizes ease over security

These might contribute to themes like:

- "Security/convenience tradeoff"
- "Perceived cognitive burden"
- "Tiered security practices"

Examples from Research

Affinity Diagramming

- Lucero, A. (2015). Using Affinity Diagrams to Evaluate Interactive Prototypes. *IFIP Conference on Human-Computer Interaction*, 231–248.

Turned thousands of messy observations from prototype testing into clear patterns by writing everything on notes and physically clustering them on walls. Teams of 2-5 researchers spent weeks moving notes around until themes emerged naturally, a tangible, collaborative way to make sense of overwhelming data.

Thematic Analysis

- Cooper, N., Horne, T., Hayes, G. R., Heldreth, C., Lahav, M., Holbrook, J., & Wilcox, L. (2022,). A Systematic Review and Thematic Analysis of Community-Collaborative Approaches to Computing Research. *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*.

Read 47 research papers about community partnerships and systematically identified patterns about power, trust, and sustainability. Shows how rigorous coding can transform a pile of papers into 7 clear themes, and reveals the tensions researchers face when working with communities.

Quality in Qualitative Research

Validity in Qualitative Research

Validity: Are your findings credible and trustworthy?

Threats to validity:

- Observer bias
- Participant reactivity
- Elite bias (studying easy-to-access people)
- Going native (losing objectivity)
- Incomplete data collection
- Cherry-picking evidence

Strategies to enhance validity:

- **Triangulation** — Multiple data sources/ methods
- **Member checking** — Validate with participants
- **Peer debriefing** — Discuss with colleagues
- **Negative case analysis** — Look for disconfirming evidence
- **Audit trail** — Document decisions
- **Rich description** — Provide detailed context

> Validity in qualitative research is about trustworthiness, not statistical proof

Reliability in Qualitative Research

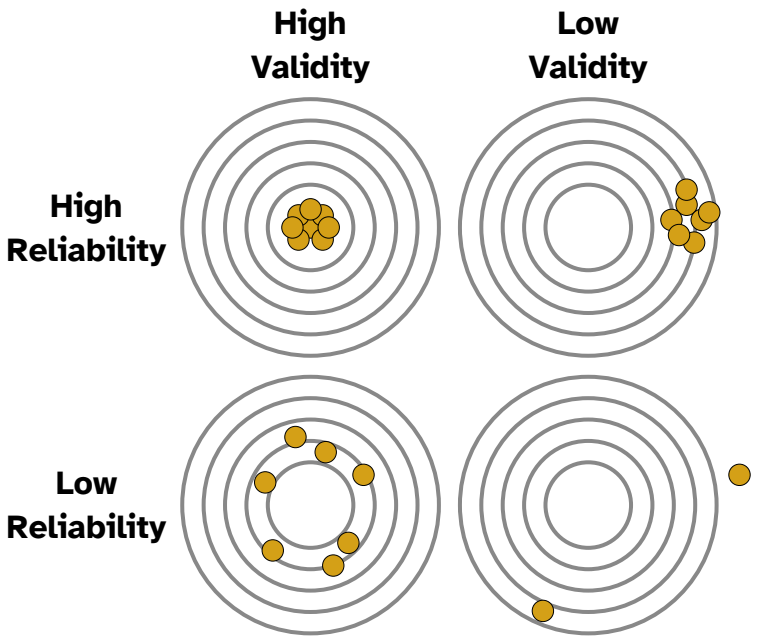
Reliability: Would another researcher reach similar conclusions?

Challenges:

- Interpretation is subjective
- Context always differs
- Researcher is the instrument

Enhancing reliability:

- Clear research protocol
- Detailed documentation of methods
- Systematic coding procedures
- Multiple coders (inter-rater reliability)
- Coding manual with definitions and examples
- Regular calibration meetings



> Document your process so others can understand and critique your work

Transparency and Open Science

Transparency: Making research process inspectable

What to share:

- Interview protocols and guides
- Coding schemes and manuals
- Analysis memos
- De-identified transcripts (with consent)
- Audit trail of decisions

Benefits:

- Increases credibility
- Enables scrutiny and critique
- Supports learning from others
- Facilitates replication/extension

> Current HCI research often lacks transparency — be part of the solution!

Ethical Considerations

1. **Informed consent**
   - Explain purpose, methods, risks
   - Right to withdraw anytime
   - Right to review transcripts
2. **Privacy and confidentiality**
   - Anonymize or pseudonymize data
   - Secure storage
   - Be careful with quotes (might identify people)
3. **Power dynamics**
   - Be aware of your position relative to participants
   - Don't exploit vulnerable populations
   - Fair compensation for time
4. **Representation**
   - Represent participants fairly and accurately
   - Avoid stereotyping
   - Consider impact of publication

> Ethics is not just about formal approval — it's an ongoing commitment throughout research

Common Pitfalls to Avoid

1. **Confirmation bias** — Only seeing what you expect
2. **Going native** — Losing critical perspective
3. **Superficial analysis** — Stopping at description, not interpretation
4. **Quote mining** — Cherry-picking quotes that support your argument
5. **Over-generalization** — Claims beyond your data
6. **Inadequate evidence** — Themes without sufficient support
7. **Ignoring negative cases** — Not addressing contradictions
8. **Unclear methods** — Can't tell how you got your findings

> Be systematic, reflective, and honest about limitations

Reporting Qualitative Findings

1. **Introduction** — Research question, why qualitative approach
2. **Related work** — Situate your study
3. **Methods** — Participants, procedures, analysis approach (be detailed!)
4. **Findings** — Themes with evidence (quotes, examples)
5. **Discussion** — Interpretation, implications, connections to theory
6. **Limitations** — Be upfront about constraints

> Your writing should bring the reader into participants' world

Qualitative + Quantitative = Powerful

Mixed methods combine qualitative and quantitative approaches

Common patterns:

- **Sequential**: Qualitative → Quantitative (explore, then measure)
- **Sequential**: Quantitative → Qualitative (measure, then explain)
- **Concurrent**: Both at once (triangulate findings)

Example: Understanding mobile app abandonment

- **Quantitative**: Analytics show 60% abandon after 1st use
- **Qualitative**: Interviews reveal why — confusing onboarding
- **Result**: Numbers identify the problem, interviews explain it

> Use the right method for each research question — don't artificially limit yourself

Key Takeaways

1. **Qualitative research provides depth and understanding** — Reveals the “why” behind behaviors and experiences
2. **Multiple methods available** — Interviews, observations, ethnography — choose based on research questions
3. **Semi-structured interviews are most common** — Balance flexibility with structure
4. **Contextual inquiry combines interview + observation** — Powerful for understanding practices in context
5. **Analysis is systematic, not ad-hoc** — Thematic analysis and affinity diagramming provide rigorous approaches
6. **Coding is fundamental** — Label data segments to identify patterns and themes
7. **Quality matters** — Attend to validity, reliability, transparency, and ethics
8. **Document your process** — Enables critique, learning, and replication
9. **Mixed methods are powerful** — Combine qualitative and quantitative for richer insights

> Qualitative research gives voice to users and builds deep understanding of their world

Resources

Essential reading:

- Lazar, J., Feng, J. H., & Hochheiser, H. (2017). **Research Methods in Human-Computer Interaction** (2nd ed.). Morgan Kaufmann.
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. **Qualitative Research in Psychology**, 3(2), 77-101.
- Beyer, H., & Holtzblatt, K. (1997). **Contextual Design: Defining Customer-Centered Systems**. Morgan Kaufmann.

Practical guides:

- Saldaña, J. (2015). **The Coding Manual for Qualitative Researchers** (3rd ed.). SAGE.
- Silverman, D. (2020). **Doing Qualitative Research** (5th ed.). SAGE.
- Emerson, R. M., Fretz, R. I., & Shaw, L. L. (2011). **Writing Ethnographic Fieldnotes** (2nd ed.). University of Chicago Press.

Tools:

- NVivo, MAXQDA, Atlas.ti (qualitative analysis software)
- Otter.ai, Trint (transcription services)

Questions?